

Technics SX-WSA1R

System Exclusive Reference

Acoustic Modeling Synthesizer Module

Operating system version 2.0

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Chapter 1

Overview

The Technics SX-WSA1R exchanges data over MIDI System Exclusive. This reference describes the messages it sends and accepts: their byte layout, the bulk-dump transfer protocol, the universal messages it honours, and what it reports when a transfer fails.

1.1 Applicability

This reference describes operating system version **2.0**. An earlier version 1 exists; its behaviour may differ and is not covered.

1.2 Ports

The instrument has two sets of MIDI sockets, MIDI 1 and MIDI 2. Ordinary performance data is handled on both. **Bulk dump transfers use MIDI 1 only** — a dump offered on MIDI 2 is not received, and the instrument sends dumps only from MIDI 1.

1.3 Verification

The message layouts, the payload encoding and the checksum rule in this reference have been checked against a bulk dump captured from a real instrument: 2883 messages carrying a SOUND and COMBINATION transfer. Of these, 2876 carry a checksum, and all 2876 verify under the rule given in chapter 3. Every transfer in the capture delivered exactly the byte count its header declared.

1.4 Conventions

Byte values are hexadecimal. A field written *nn* takes one of the values listed for it. Ellipses inside a message stand for a body whose length depends on the message.

Every message begins with F0 and ends with F7, as required by MIDI.

Real-time messages are safe inside a transfer

System Real Time bytes (F8–FF) may be interleaved anywhere inside a System Exclusive message without disturbing it. Conversely, while the instrument is receiving a System Exclusive message it ignores F8 TIMING CLOCK, FA START, FB CONTINUE and FC STOP for transport purposes.

Chapter 2

Message framing

2.1 Identifiers

The instrument accepts exactly two System Exclusive identifiers:

identifier	meaning
50	Matsushita (Technics)
7E	Universal Non-Real Time

A message carrying any other identifier is tracked to its F7 and discarded. No error is reported for it.

There is no device or unit number after the identifier

Many instruments place a device id immediately after the manufacturer identifier. The SX-WSA1R does not. The byte following 50 is already the message-family byte described in chapter 3. A unit field does appear later in some families, but it is not where the convention would put it.

2.2 Termination

A message ends at F7. If any other status byte in the range 80–F6 arrives while a message is open, the message is abandoned and its contents discarded.

2.3 Length limit

A complete message may not exceed 256 bytes counting F0 and F7, that is 254 bytes between them. A longer message is abandoned and, inside a bulk-dump session, reported as a reception error. Dumps sent by the instrument use the full 256 bytes for all but the last message of each transfer.

Chapter 3

The Technics message set

3.1 Message families

After F0 50, the third byte selects a message family. The instrument accepts fourteen values:

21 22 23 24 25 27 28 29 2A 2B 2C 2D 7E 7F

26 is not accepted. The SX-WSA1R's own product identifier is 2D; its outgoing bulk-dump data messages all begin F0 50 2D.

3.2 The model triple

The families 21, 22, 2B, 2C and 2D continue with three fixed bytes:

04 *nn* 11

where *nn* is 00 or 01. A message whose triple the instrument does not recognise is rejected with ERROR 40.

The middle byte is not a constant of the product

The instrument transmits 04 00 11 and accepts both 04 00 11 and 04 01 11. A bulk dump captured from a real instrument carries 04 01 11 throughout. The transmitted value is fixed — it is not derived from the keyboard/rack setting described in chapter 7 — so the middle byte does not distinguish the two models either. A receiver must accept both values and must not use the middle byte to identify anything.

3.3 Short messages

These messages are complete as shown. They form the control vocabulary of a bulk-dump session.

message	meaning
F0 50 23 7E F7	acknowledge
F0 50 24 7E F7	not acknowledged
F0 50 27 7E F7	end of category
F0 50 28 7E F7	end of dump
F0 50 29 7E F7	abort
F0 50 2A 7E F7	destination full
F0 50 21 04 <i>nn</i> 11 F7	enquiry
F0 50 22 04 <i>nn</i> 11 F7	enquiry reply

3.4 Tempo

The family 25 carries a tempo setting, in the range 40 to 300 beats per minute. A value outside that range is treated as 120.

See chapter 7 — the tempo message is one of two things the rack does not support.

3.5 Parameter families

The families 2B and 2C carry individual parameter traffic rather than bulk transfer. Each addresses a large parameter space, and neither is enumerated in this edition; see chapter 9.

Chapter 4

Bulk dump

Bulk dump transfers the instrument's internal data as System Exclusive. The instrument is silent for the duration, and the transfer uses the MIDI 1 sockets only.

4.1 Categories and what each transmits

The SYSEX BULK DUMP screen offers five categories. Each sends one or more data messages at fixed dump addresses.

category	address bytes	length bytes	length	blocks
SYSTEM, PART & MIDI	40 00 00	00 00 20	32	
	40 00 20	00 12 60	2 400	
SOUND	20 00 00	10 00 00	262 144	32
COMBINATION	50 00 00	00 06 00	768	
	50 06 00	05 40 00	90 112	16
SEQUENCER	60 00 00	00 18 00	3 072	
	60 18 00	01 70 00	30 720	
	62 08 00	set at send time		
TOTAL KEYBOARD	sends the other categories in turn			

Lengths are in source bytes; on the wire each occupies twice as many, because of the payload encoding described below. The block counts are the instrument's 32 sounds and 16 combinations: those two categories repeat one block layout that many times.

The SEQUENCER category is the only one whose final message has no fixed length — it carries as much recorded material as the instrument holds.

The dump address is not a memory address

The three address bytes identify a *region of the dump format*, not a location in the instrument's memory. The values above are the only ones the instrument emits, and a receiver should treat them as fixed identifiers for the four categories rather than attempt to compute from them.

4.2 Data message

F0 50 2D 04 *nn* 11 *a2 a1 a0* *l2 l1 l0* *data...* *c* *sum* F7

nn unit field, 00 or 01.

a2 a1 a0 destination address, three bytes, most significant first, seven bits each: the value is $a_2 \times 128^2 + a_1 \times 128 + a_0$.

l2 l1 l0 length in *source* bytes, encoded the same way.

data the payload, encoded as described below.

c continuation flag: 01 if further messages follow, 00 on the last.

sum checksum.

A message is filled to at most 252 bytes before being sent; longer transfers are split.

4.3 Payload encoding

Each source byte is sent as two message bytes, **high nibble first**:

$$b \longrightarrow (b \gg 4), \quad (b \& 0F)$$

So source byte **F1** appears on the wire as **0F 01**, and the payload occupies twice as many bytes as the length field states. A receiver rejoins them as $((\text{first} \& 0F) \ll 4) | (\text{second} \& 0F)$.

An odd number of payload bytes is invalid.

4.4 Checksum

Let *S* be the eight-bit sum of every byte from the identifier **50** through the continuation flag inclusive. Then

$$\text{sum} = (0 - S) \& 7F$$

The leading **F0**, the checksum byte itself and the trailing **F7** take no part in the sum. Equivalently, the sum of every byte after **F0** including the checksum is zero modulo 128.

The checksum is not verified for every family

On reception the checksum is checked only for families **7E**, **2B**, **2C** and **2D**. For the other accepted families a wrong checksum passes unremarked.

4.5 Requesting a dump

A dump can be asked for over MIDI; the front-panel **SEND** button is not the only way to start one. The request names a category by its region byte:

F0 50 2B 04 nn 11 area 00 00 x x x F7

<i>area</i>	category requested
40	SYSTEM, PART & MIDI
20	SOUND
50	COMBINATION
60	SEQUENCER

The instrument then runs exactly the transfer the corresponding menu row would have run.

4.6 Opening a transfer

A dump begins with a two-step exchange. Each step is attempted up to three times.

1. The sender transmits F0 50 21 04 00 11 F7 and expects F0 50 22 04 *nn* 11 F7.
2. The sender transmits F0 50 22 04 00 11 F7 and expects F0 50 23 7E F7.

If the exchange succeeds, every subsequent data message must be acknowledged. If all three attempts of the first step fail, **the dump proceeds anyway, without acknowledgements**, at a fixed pace of roughly 51 ms between messages.

4.7 Acknowledgement and retry

With acknowledgements in force, each data message must be answered with F0 50 23 7E F7. Any other reply causes the same message to be retransmitted, up to three attempts, after which the transfer is abandoned.

A reply is awaited for about 5.1 seconds; during the opening exchange the wait is about 2.05 seconds.

4.8 Continuation

Data messages after the first open with F0 50 7E, followed by payload, continuation flag, checksum and F7.

4.9 Closing

At the end of each category the sender transmits F0 50 27 7E F7 and waits for an acknowledgement. At the end of the whole transfer it transmits F0 50 28 7E F7. If the transfer was abandoned after three failed retries it transmits F0 50 29 7E F7.

4.10 Receiving

When the SX-WSA1R is the receiver it answers each data message with F0 50 23 7E F7 to accept, F0 50 24 7E F7 to reject, and F0 50 2A 7E F7 when the incoming data is larger than the destination.

While a bulk-dump session is open the instrument accepts only the 50 identifier; universal messages are not honoured until the session ends.

Memory-full is treated as end-of-dump

When the SX-WSA1R is *sending* and the receiver answers F0 50 2A 7E F7 to report that its memory is full, the instrument treats that exactly as it treats F0 50 28 7E F7, end of dump. It stops, and it reports success. A sender that needs to know whether the data arrived cannot rely on the instrument to notice.

Chapter 5

Universal System Exclusive

5.1 Recognised messages

The instrument honours two universal messages, and transmits both.

message	meaning
F0 7E 7F 09 01 F7	General MIDI System On
F0 7E 7F 09 02 F7	General MIDI System Off

Turning General MIDI on restricts the sounds and operations available and may rearrange the percussion map. General MIDI System Off has no effect if General MIDI is already off.

Both messages are also written into Standard MIDI Files produced by the instrument, as a five-byte System Exclusive event at delta time zero.

5.2 Messages not recognised

A controller expecting any of the following will receive no response:

message	
F0 7F 7F 04 01 <i>ll mm</i> F7	Master Volume
F0 7E <i>dd</i> 06 01 F7	Identity Request
F0 7E 7F 09 03 F7	General MIDI 2 System On

The instrument does not implement Identity Reply, so it cannot be auto-detected by a controller that relies on it. The entire Universal Real Time identifier 7F is unrecognised.

Chapter 6

Errors

Transfer failures are reported on the SYSEX BULK DUMP screen. A System Exclusive message arriving while any other screen is displayed produces no visible error.

6.1 The messages

screen	meaning
COMPLETED!	the transfer finished
ERROR 40!	the message is addressed to a different product
ERROR 41!	reception failed
ERROR 42!	transmission failed
ERROR 21!	the destination is full

Not every failure is a 40, 41 or 42

When incoming data is larger than the destination area, the instrument reports ERROR 21! MEMORY FULL. Software that recognises only the three documented System Exclusive errors will mis-read a truncated restore as something else.

6.2 ERROR 40 — wrong identification

Raised when the message family byte is accepted but the bytes that follow it do not match the product. For a message addressed to the SX-WSA1R this means the model triple was not 04, then 00 or 01, then 11.

6.3 ERROR 41 — reception

The general reception failure. It covers:

- a message inside a session that does not begin F0 50;
- a status byte other than F7 arriving inside an open message;
- a framing, parity or overrun error on the MIDI input, or loss of active sensing;
- no reply within the timeout;
- a message longer than 254 bytes;

- an odd number of payload bytes;
- a continuation flag that is neither 00 nor 01;
- a checksum mismatch.

6.4 ERROR 42 — transmission

Raised when the instrument cannot complete an outgoing transfer internally, roughly a second after the attempt stalls. In this version it has a single cause and does not indicate a fault on the MIDI cable or at the receiving device.

6.5 Interference from other traffic

While a bulk-dump session is open, unrelated System Exclusive traffic arriving on the MIDI input can set the error state, and will then be reported when the session ends. Keep other sources off the MIDI 1 input for the duration of a transfer.

Chapter 7

SX-WSA1 and SX-WSA1R

The SX-WSA1 keyboard and the SX-WSA1R rack module run the same program and speak the same System Exclusive protocol. Two features are nonetheless available on the keyboard and not on the rack, in both directions.

	SX-WSA1 (keyboard)	SX-WSA1R (rack)
message framing, checksum, handshake	yes	yes
SOUND bulk dump	yes	yes
COMBINATION bulk dump	yes	yes
SYSTEM, PART & MIDI bulk dump	yes	yes
dump requests	yes	yes
General MIDI On / Off	yes	yes
SEQUENCER bulk dump	yes	no
tempo message (family 25)	yes	no

On the rack, a SEQUENCER dump sends nothing and says nothing

The SEQUENCER row is present on the rack's SYSEX BULK DUMP menu, and selecting it and pressing SEND appears to work. Nothing is transmitted — not even the end-of-category message that closes every other transfer. A librarian waiting for data will simply time out.

The same applies to the SEQUENCER portion of TOTAL KEYBOARD.

The tempo message is refused in both directions on the rack: it will neither act on one received nor send one.

The model triple does not tell you which you are talking to

Both models transmit the same model triple, so it cannot be used to distinguish them. The only reliable test over MIDI is behavioural — request a SEQUENCER dump and see whether anything arrives.

Chapter 8

Relationship to other Technics instruments

The message engine described in this reference is not unique to the SX-WSA1R. At least two other Technics instruments implement the same protocol: the SX-KN5000 and the SX-KN1500.

8.1 What is shared

All three use the 50 identifier, the same message families, the same short control messages, the same payload encoding and the same checksum rule. The fixed messages are byte-for-byte identical:

```
F0 50 23 7E F7  F0 50 24 7E F7  F0 50 27 7E F7  F0 50 28 7E F7  F0 50 29 7E
F7  F0 50 2A 7E F7
```

The error mapping is also nearly common: of the thirty-four status values, thirty-two produce the same report on the SX-WSA1R and the SX-KN5000.

8.2 What distinguishes the products

The model triple is the discriminator. A message whose triple does not match the receiving instrument is refused with ERROR 40.

instrument	model triple
SX-WSA1R	04 00 11, 04 01 11
SX-KN5000	01 28 12
SX-KN1500	01 24 11, 01 25 11, 01 26 11

A librarian that supports more than one of these instruments needs only to vary the triple and the dump region addresses; the framing, handshake, encoding and checksum are common code.

8.3 Dump regions differ

The dump region addresses are per-product. The SX-KN5000 and SX-KN1500 share eleven of their thirteen region and length pairs with each other, but the SX-WSA1R shares none of its eight with either — its categories are its own.

The SX-WSA1R does not accept the GS-style messages its siblings do

The SX-KN5000 and SX-KN1500 additionally accept a small set of messages in the 42 12 parameter form, including the sequence 42 12 40 00 7F 00 41 used elsewhere in the industry as a reset. **The SX-WSA1R accepts none of them.** Software that resets a KN-series instrument that way and assumes the SX-WSA1R will follow will find it does not respond.

Chapter 9

Scope of this edition

This reference covers message framing, the bulk-dump transfer protocol, the universal messages, and error reporting. The following are not covered.

Parameter addresses. The message families 2B and 2C carry individual parameter traffic and each addresses a large parameter space. The addresses themselves are not tabulated here.

Block contents. Chapter 4 gives the address, length and repeat count of every message a dump contains, but not the meaning of the bytes *inside* a block — the structure of a sound, a combination or a sequence.

Operating system version 1. Everything here describes version 2.0.